

VENGATESHWARAN N

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SUMMARY

Principal AI Systems Architect & Software Engineer specializing in Computational Intelligence, Large Language Models (LLMs), and Scalable Cloud Infrastructure. Expert in engineering Production-Ready MLOps Pipelines, Distributed Systems, and Neural Architectures with a focus on bridging Mathematical Research with Enterprise Software Engineering to deliver high-performance Artificial Intelligence solutions.

TECHNICAL SKILLS

AI & Machine Learning: Supervised Learning, Unsupervised Learning, Reinforcement Learning, Neural Networks, Deep Learning (DL), Natural Language Processing (NLP), Computer Vision (CV), Generative AI (GenAI), Transformers (BERT, GPT).

Data Science & Analytics: Exploratory Data Analysis (EDA), Statistical Modeling, Bayesian Inference, Feature Engineering, Regression, Classification, Clustering, Time-Series Analysis.

Programming & Engineering: Python (NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow), PHP (CodeIgniter 4, Laravel), SQL (MySQL, PostgreSQL), JavaScript, RESTful API Design, Microservices, MVC Architecture.

Tools & DevOps: MLOps, Git, CI/CD, Docker, Kubernetes, Linux, Power BI, Tableau, Advanced Excel, MATLAB, R Programming.

EXPERIENCE

Full Stack Web Developer – NPR College of Education

Jul 2025 – Present

- Architecting **Enterprise Software Solutions** using Python, PHP 8.x, and MySQL; optimized backend system architecture to support **5,000+ concurrent academic users**.
- Engineered **Relational Database Management Systems (RDBMS)** with RBAC and cryptographic protocols, reducing potential security vulnerabilities by **95%**.
- Implementing **MLOps Frameworks** for AI model deployment, accelerating data pipeline processing speeds by **40%** within the SDLC.

Data Scientist & AI Researcher (Intern) – Oasis Infobyte (Remote)

Oct 2023 – Nov 2023

- Synthesized **End-to-End Machine Learning Pipelines** using Scikit-learn and NumPy; achieved **92% accuracy** in multi-class statistical classification tasks.
- Architected **Natural Language Processing (NLP)** systems for sentiment analysis, leveraging hyperparameter optimization to increase model F1-score by **15%**.

Machine Learning Engineer (Intern) – CodSoft (Remote)

Sep 2023 – Oct 2023

- Deployed **Anomaly Detection Algorithms** using XGBoost and GBDT, identifying fraudulent transactions with a **98% detection rate**.
- Leveraged **Python** for PCA and stochastic modeling, reducing computational latency by **35%** while maintaining high model generalizability.

TECHNICAL PROJECTS

IGC-Mail: Enterprise Workflow & Communication Engine (PHP, SQL, MVC)

- Architected a **Scalable Full-Stack System** for institutional data governance, reducing inter-departmental communication latency by **50%**.
- Engineered **JSON-driven Logic** and Shield authentication to secure sensitive governance data for **10+ administrative departments**.

IGC Event Management: Institutional Workflow Automation System (CI4, PHP, MySQL)

- Designed an **End-to-End Governance Workflow** that automated **100%** of institutional activity approvals and auditing stages.
- Developed **Dynamic Dependency Logic** ensuring zero coordination errors across **6 prep units** (Transport, Hospitality, etc.) before event commencement.
- Implemented **Transactional Resource Allocation** and compliance timelines, cutting administrative planning time by **30%**.

CUBY.AI: Neural Automation Agent (Python, NLP, GenAI)

- Developed an **Intelligent AI Agent** integrating NLU and GenAI, successfully automating **25+ OS-level routine tasks**.
- Implemented **STT** and intent recognition workflows, resulting in a **60% increase** in human-computer interaction (HCI) efficiency.

Computer Vision Engine: Image Segmentation & CV (OpenCV, NumPy)

- Built a **Production-Ready CV Application** for pixel-level segmentation, achieving high-fidelity results comparable to **professional editing suites**.
- Optimized **Batch Processing Data Pipelines** with multithreading, allowing the concurrent processing of **100+ images per minute** on localized hardware.

EDUCATION

Master of Computer Applications (MCA) | AI & Systems Design – Bharathidasan University (2026)

Bachelor of Science in Computer Science (B.Sc.) | Data Science & Algorithms – Madurai Kamaraj University (2024) | Aggregate: 80%

CERTIFICATIONS

- **Machine Learning & Data Science:** Predictive Analytics, Ensemble Learning, and Python Frameworks — Great Learning.
- **Deep Learning & Neural Networks:** CNN, RNN, Transformers, and AI Optimization — Infosys Springboard.
- **Data Analytics & BI:** Microsoft Power BI, Tableau, and Data Visualization.

LANGUAGES

Tamil (Native), English (Fluent), Kannada (Intermediate)

INTERESTS

Artificial Intelligence Systems, Intelligent Automation, Data Analytics, Continuous Learning